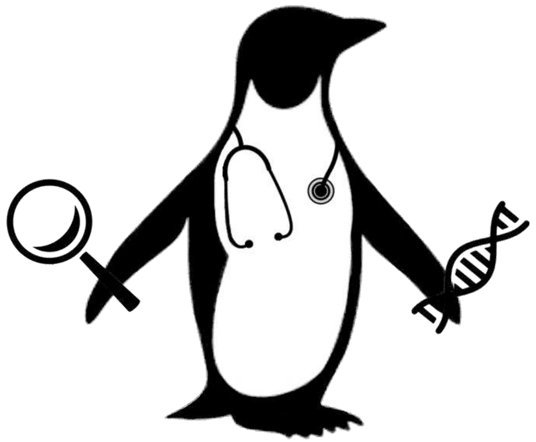


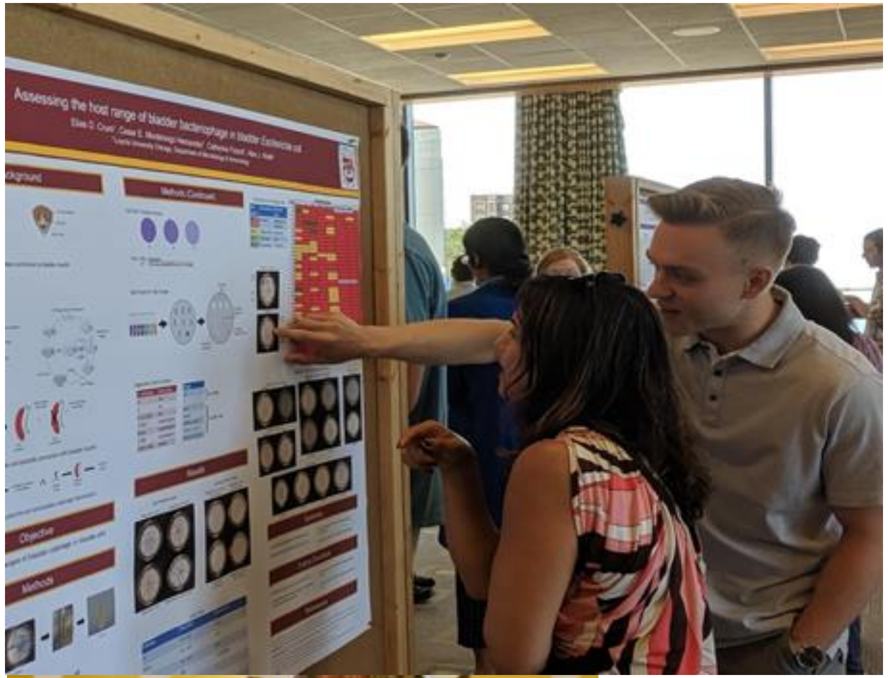
PENGQUIN:

PErsoNal Genome QUery IN healthcare and clinical practice

& other RDF-related fun

Elias Crum

SwissProt Presentation
March 25th, 2025



Personal Genome Data Usage?

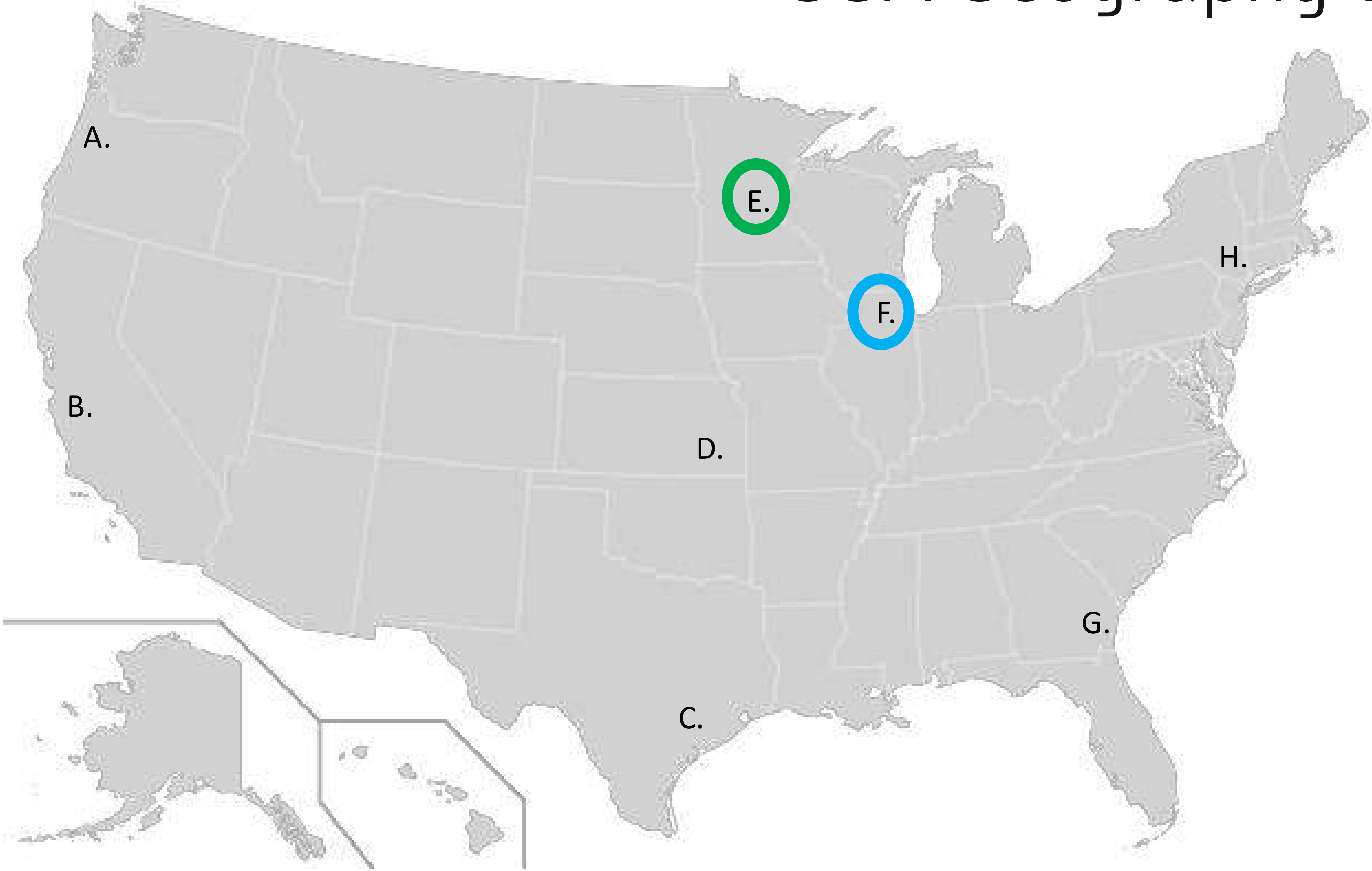
My Journey:

BSc + MSc - Bioinformatics

Prokaryotic Genomics

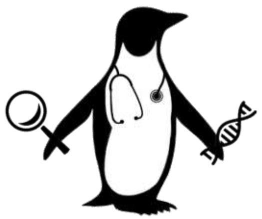
How can we increase **genomics** use in **medicine**?

USA Geography Lesson

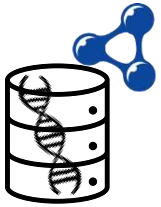


PENGQUIN

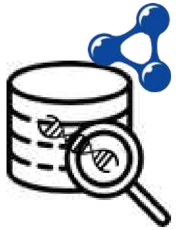
Genomic Data



Clinical Care



Semantic, Linked Data



Query Capabilities

GENOME Pods

Framework + App

Promoters: Dr. Gökhan Ertaylan[1], Dr. Bart Buelens[1], Dr. Ruben Taelman[2], Prof. Dr. Ruben Verborgh[2]

[1] = VITO NV
[2] = Ghent University

PENGQUIN Goals

Improve genomic data USAGE and SCALABILITY

Improve genomic data store accessibility (while maintaining privacy)

Represent genomic data semantically as Linked Data

Query genomic data and linkages

Beacon

Connect to existing initiatives / applications

Proof-of-concept

Genome Pods

Develop a framework and web application

Talk Outline

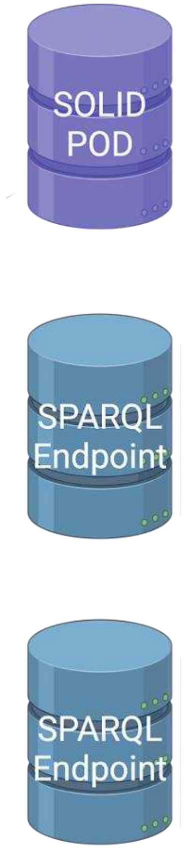
Decentralized Data Storage *(CHIST-ERA TRIPLE Project)*

- Introduction to the Project
- Introduction to Solid Pods
- The Solid Cockpit web application (and maybe a use-case example if we have time)

Federated Querying *(Real World Federated Query Survey)*

- Motivation + Challenges
- Progress and preliminary findings
- Future aims

CHIST-ERA TRIPLE Project



Standardized documentation and resource statistics

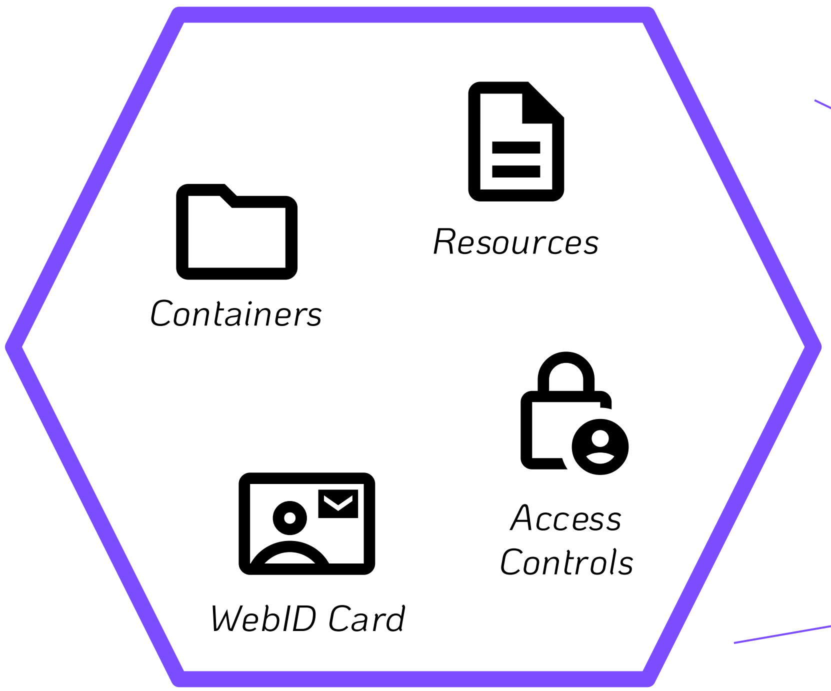
chist-era

What is a Solid Pod?

Secure, Decentralized Data Storage Specification

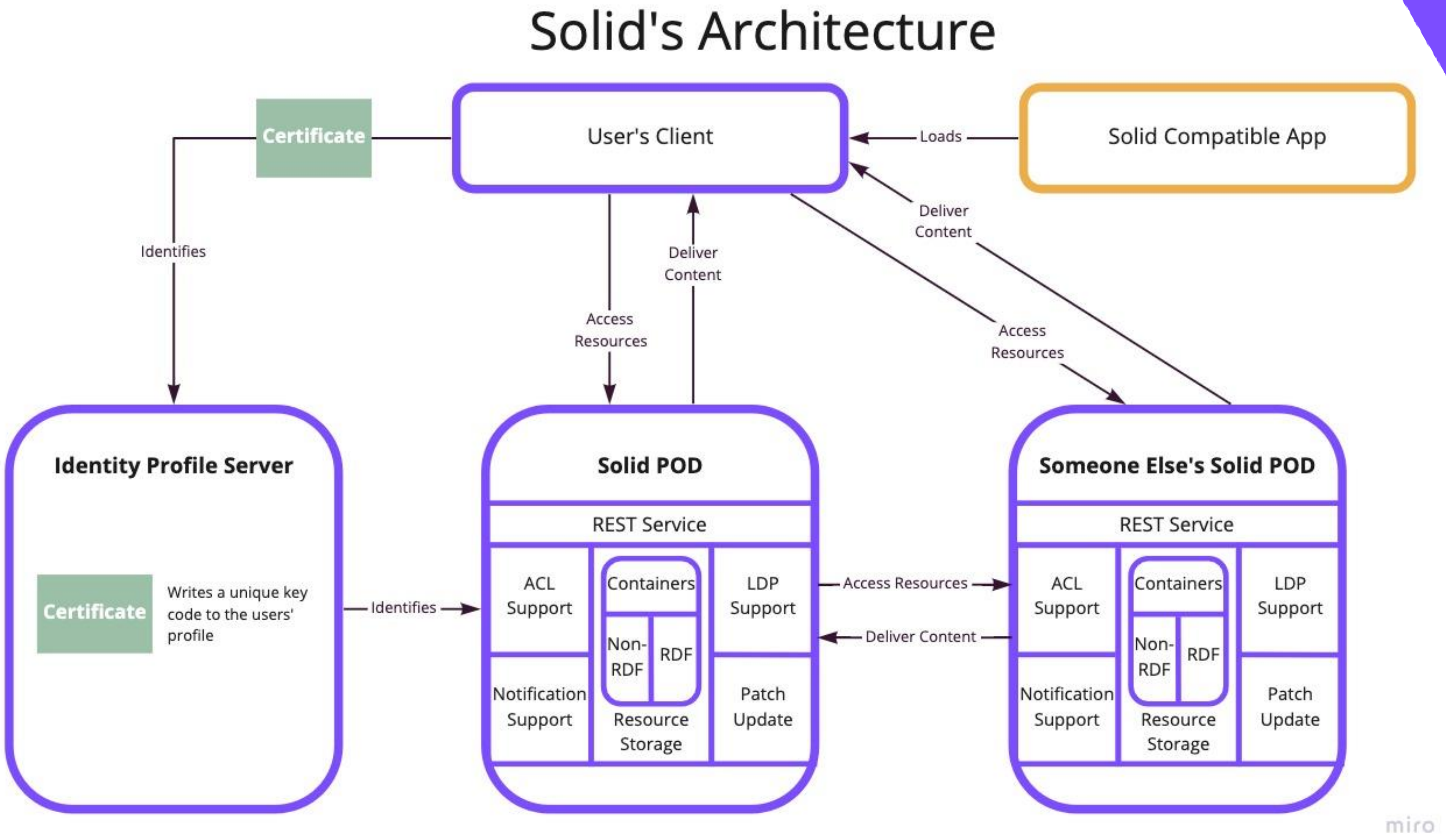
- ↳ *Pod exists on a Server (accessible to the Web)*
- ↳ *Can store files, RDF data, and Linked Data*
- ↳ *Provides control policies for stored data*

Elias' Pod



https://solidproject.org/about

What is a Solid Pod? - *Detailed*



https://solid.georgetown.domains/technical-component/

Solid Pod vs SPARQL Endpoint?

Solid Pod

+ decentralized storage strategy
+ can store both file data and RDF data
+ granular privacy policy capabilities

- native querying capabilities limited
- not designed for large-scale data
- no computation attached to data store

SPARQL Endpoint

- centralized storage strategy
- triple store (only RDF)
- limited privacy capabilities

+ designed for high-performance querying
+ excels with large datasets
+ computation directly attached to data store

How to query a Solid Pod?

A Query Engine

COMUNICA

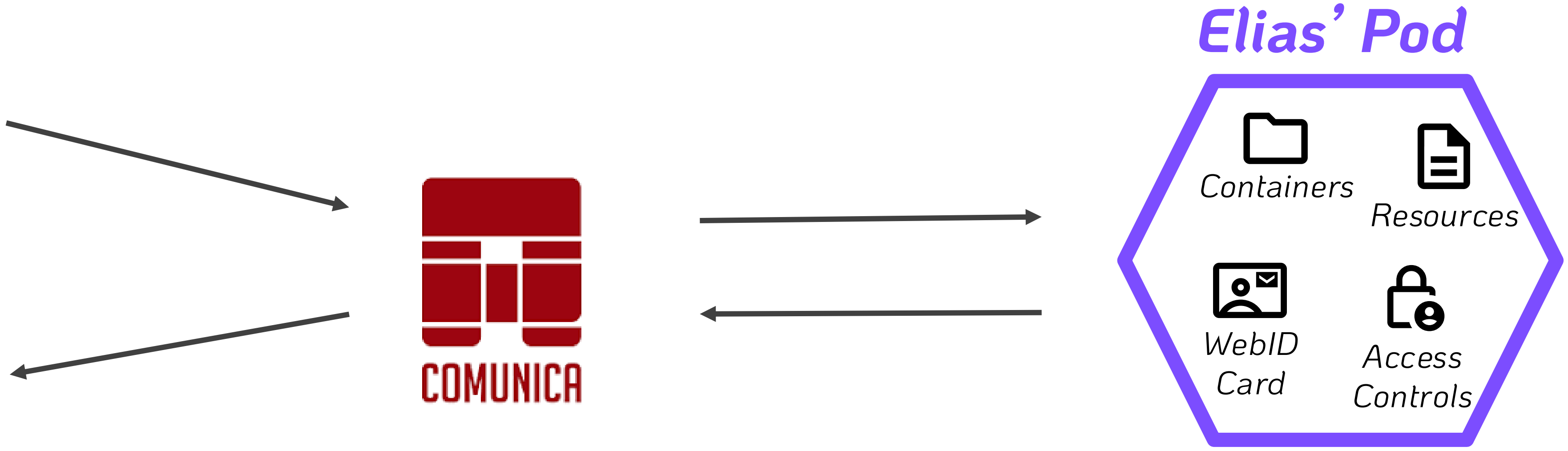
- Query computation client-side
- Allows for SPARQL query execution over Solid Pod API
- Linked Data traversal + heterogeneous result joining capabilities

SPARQL Query

```
SELECT DISTINCT ?p {
    ?s ?p ?o
} LIMIT 10
```

Results Binding Stream

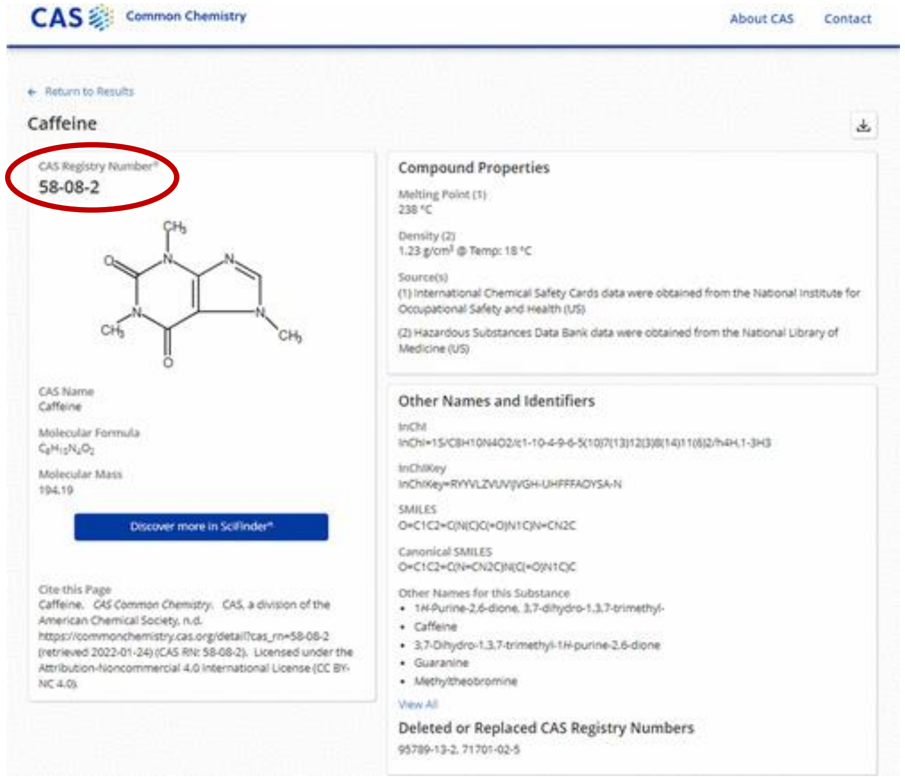
SPARQL+JSON



https://comunica.dev/about

Solid Pod Querying – *Practical Example Concept*

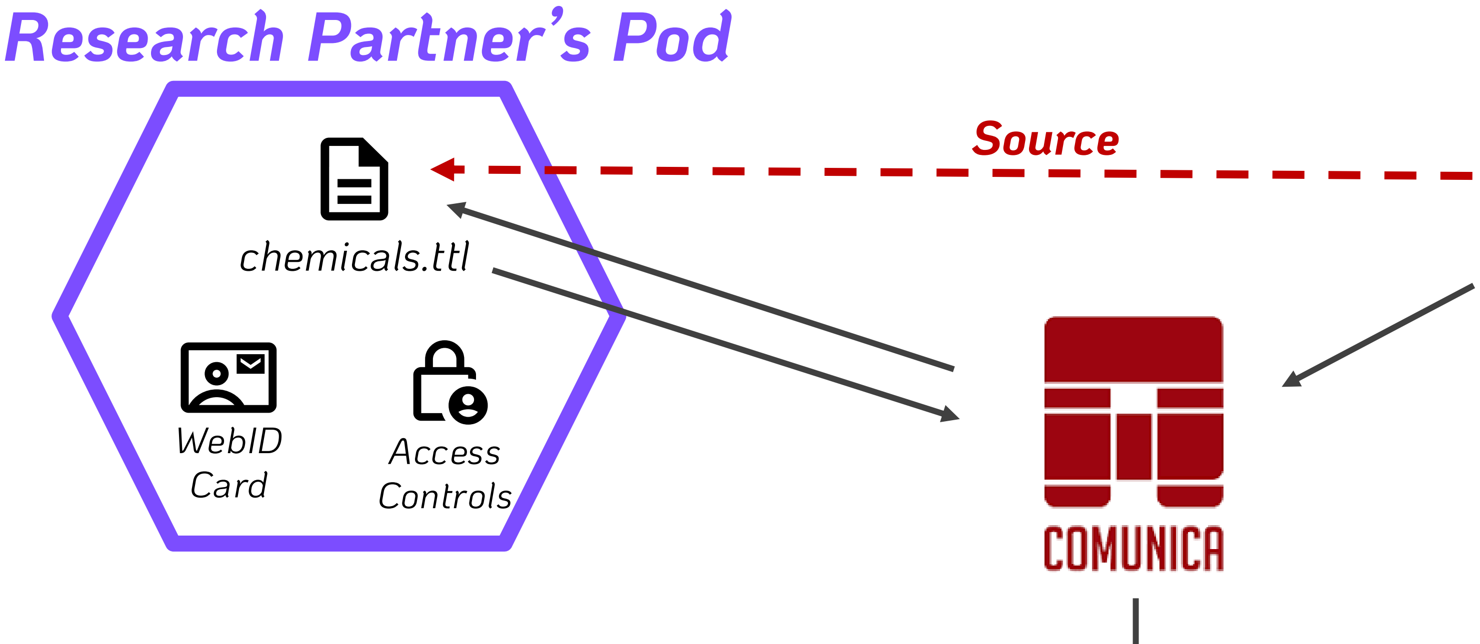
1. A research partner has a [proprietary] list of CAS numbers from biological sample analysis…



2. We want to find other chemicals with similar molecular structures for a toxicity study

3. How could we design and execute a SPARQL query to achieve this goal?

Solid Pod Querying – *Practical Example*



SPARQL Query

```
prefix sio: <http://semanticscience.org/resource/>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX sio: <http://semanticscience.org/resource/>
PREFIX vocab: <http://rdf.ncbi.nlm.nih.gov/pubchem/vocabulary#>
PREFIX sachem: <http://bioinfo.uochb.cas.cz/rdf/v1.0/sachem#>
PREFIX endpoint: <https://idsm.elixir-czech.cz/sparql/endpoint/>

SELECT ?CAS ?COMPOUND ?SIMILAR_COMPOUND ?SCORE WHERE {
  ?s sio:SIO_000300 ?CAS .

  SERVICE endpoint:idsm {
    ?SYNONYM a sio:CHEMINF_000446. # CAS registry number
    ?SYNONYM sio:SIO_000300 ?CAS.
    ?SYNONYM sio:SIO_000011 ?COMPOUND.
    ?COMPOUND a vocab:Compound.
    ?ATTRIBUTE a sio:SIO_011120. # molecular structure file
    ?ATTRIBUTE sio:SIO_000011 ?COMPOUND.
    ?ATTRIBUTE sio:SIO_000300 ?MOLFILE.
    {
      SERVICE endpoint:chebi {
        [ sachem:compound ?SIMILAR_COMPOUND;
          sachem:score ?SCORE ] sachem:similaritySearch [
            sachem:query ?MOLFILE;
            sachem:cutoff "0.7"^^xsd:double;
            sachem:similarityRadius '3'^^xsd:integer;
            sachem:aromaticityMode sachem:aromaticityDetectIfMissing;
            sachem:tautomerMode sachem:inchiTautomers
        ].
      }
    }
    UNION
    {
      SERVICE endpoint:chebi {
        ?SIMILAR_COMPOUND sachem:substructureSearch [ sachem:query '[As]' ].
      }
    }
  }
}
ORDER BY DESC(?SCORE)
```

Results

	CAS	COMPOUND	SIMILAR_COMPOUND	SCORE
0	92-67-1	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_1784	1.0
1	34256-82-1	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_2394	1.0
2	485-31-4	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_83371	1.0
3	485-31-4	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_83367	1.0
4	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_4034	1.0
5	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_39313	1.0
6	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_39309	1.0
7	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_39310	1.0
8	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_39312	1.0
9	121-14-2	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_920	1.0

My Contributions

Solid Cockpit

knowledgeonwebscale.github.io/solid-cockpit/

My Contributions (specifics)

knowledgeonwebscale.github.io/solid-cockpit/

A Web Application to:

- *Upload Data to a Solid Pod*
- *Browse contents of a Solid Pod*
- *Edit Privacy of (and share) data in a Solid Pod*
- *Query Data in Solid Pods + SPARQL Endpoints*
  - ↳ *Implement a query cache for storing queries and results*

Real World Federated Query Survey - *Intro*

Motivation:

Federated querying for answering questions over **multiple heterogeneous data sources**

Performance is currently ~variable~

We want to detail *WHY* performance varies so much && suggestions for improvement

Assess federation algorithms on REAL endpoints

FedX – uses `ASK` queries to inform source-selection

SPLENDID – uses `VoID` descriptions to inform source-selection

Real World Federated Query Survey *- Design*

SPARQL Endpoints:

“Real World” Federated Queries: https://sib-swiss.github.io/sparql-examples/

SPARQL Query Engine: https://comunica.dev/

Real World Federated Query Survey – *[Preliminary] Findings*

Preliminary Results (endpoints and queries):

Federated querying is not ”easy”

↳ Errors related to SPARQL endpoint availability

Invalid SPARQL endpoint response from https://semantic.eea.europa.eu/sparql (HTTP status 404)

↳ Errors caused by SPARQL endpoint-related errors

Invalid SPARQL endpoint response from https://sparql.orthodb.org/sparql (HTTP status 500)

↳ Errors related to SPARQL endpoint rate-limiting

Invalid SPARQL endpoint response from https://query.wikidata.org/sparql (HTTP status 429): Rate limit exceeded

↳ Errors related to SPARQL endpoint time-outs

Unexpected ""!"" at position 0 in state STOP

Real World Federated Query Survey – *[Preliminary] Findings*

Preliminary Results (algorithms):

Federated querying is not ”easy”

↳ Errors within Comunica

Actor urn:comunica:default:query-operation/actors#source requires an operation with source annotation.

fetch failed

Could not retrieve https://idsm.elixir-czech.cz/sparql/endpoint/chebi (HTTP status 400)

FATAL ERROR: Ineffective mark-compacts near heap limit Allocation failed - JavaScript heap out of memory

↳ Errors related to algorithms

Actor urn:comunica:default:/actors#rdf-join cannot complete join with Asym-hash actor.

Real World Federated Query Survey – *Future Work*

Future Aims:

Execute 1 month worth of queries

↳ 4 experimental runs per week

Vary time + part of week experiment is performed

Publication goals:

- In-Use Paper
- Journal Paper Extension (benchmark creation)

Algorithmic variation

- `VoID` vs `ASK` queries
- Various join algorithm combinations
- Rate-limiting
- Service-clause removal

Metrics Assessed

- Total execution time
- Query planning time
- Result set consistency
- # SPARQL requests
- # HTTP requests
- Individual query arrival times

For more detail see: https://docs.google.com/document/d/1U-RtYsJw9raw4U5Qe6BHfSh2VYGnUIkbT9-aSyJionU/edit?usp=sharing

Thank you!

More PhD specifics

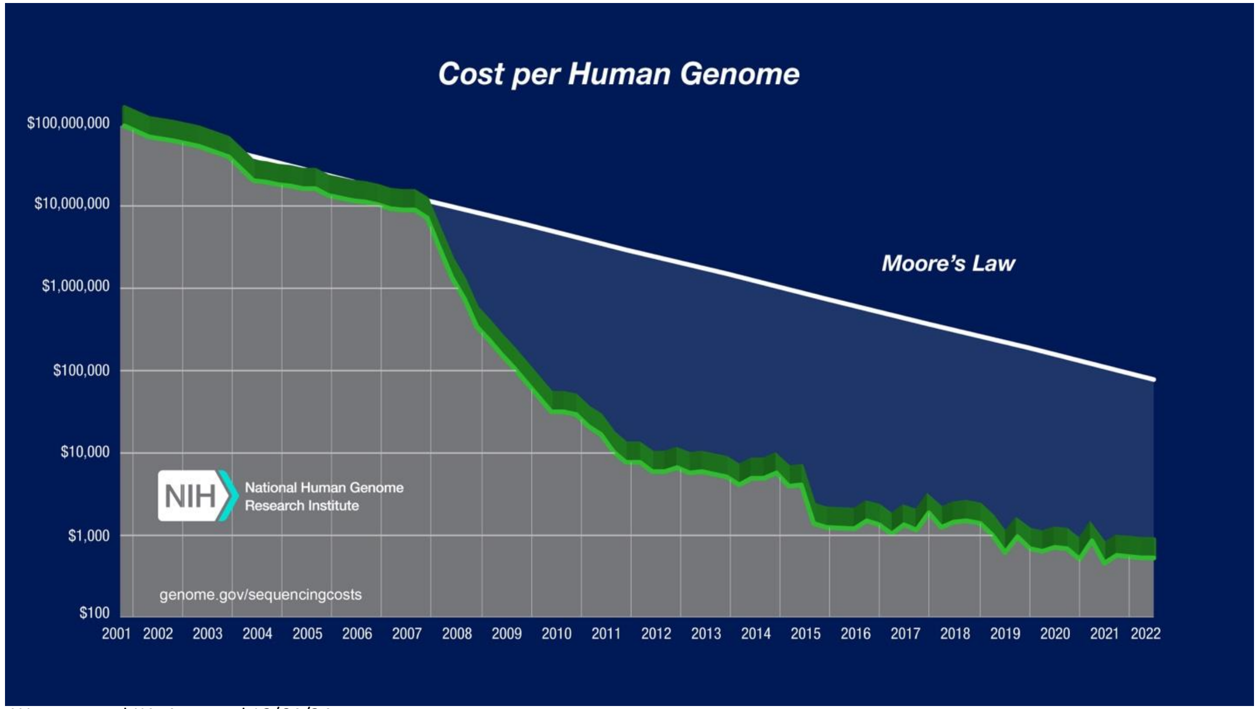
LinkedIn

GitHub

Supplemental Slides

Genome Sequencing in Health Care

Sequence Generation Costs



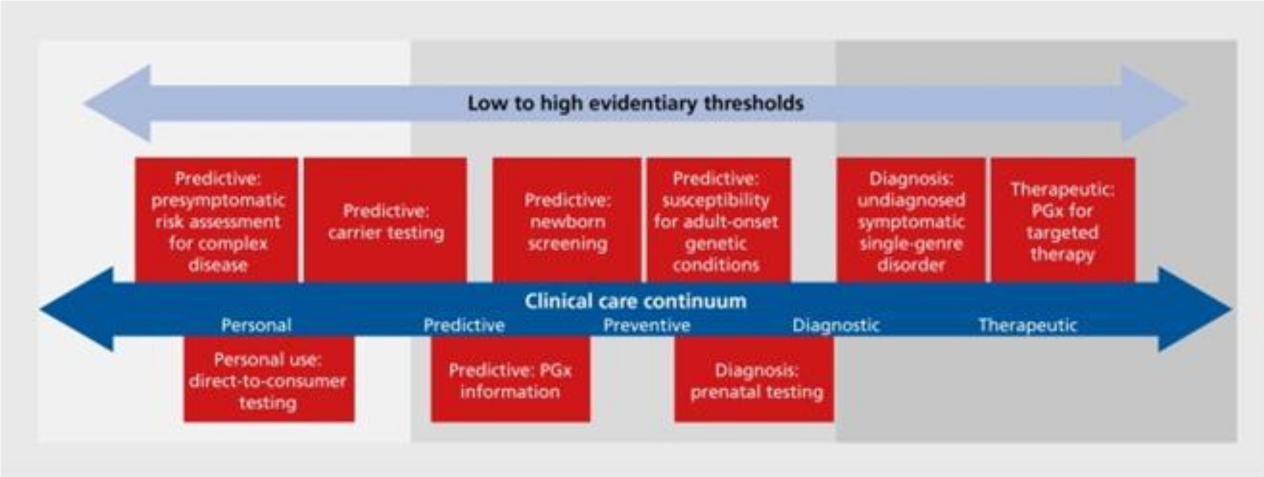
Wetterstrand KA. Accessed 12/04/24.

Currently getting close to $100 per genome[4]

Per-base cost of sequencing has been dropping by half every ~5 months

Per-base cost of sequence data storage is dropping by half every ~14 months

Use-cases



10.31887/DCNS.2016.18.3/jkrier

Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

Scaling Genome Data Usage

Research genomic data sharing initiatives:

FHIR

Beacon

OHDSI
OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

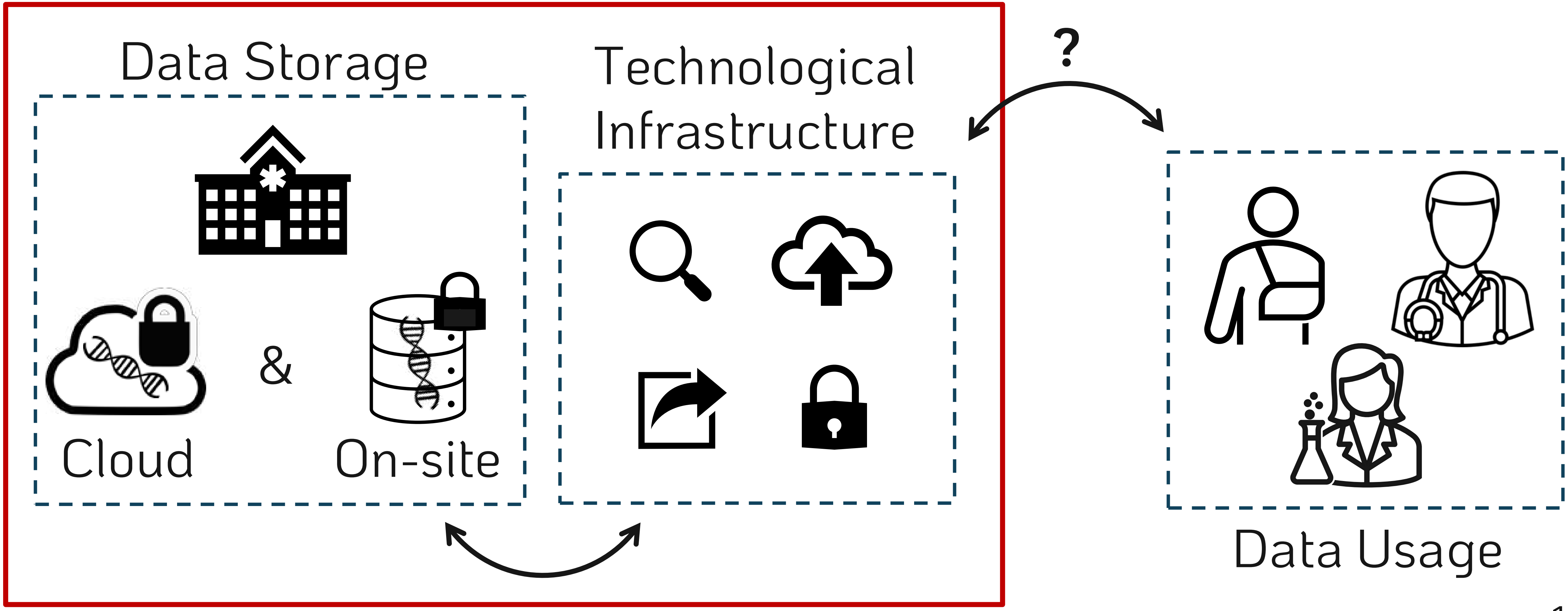
State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US)
EU Health Data Spaces

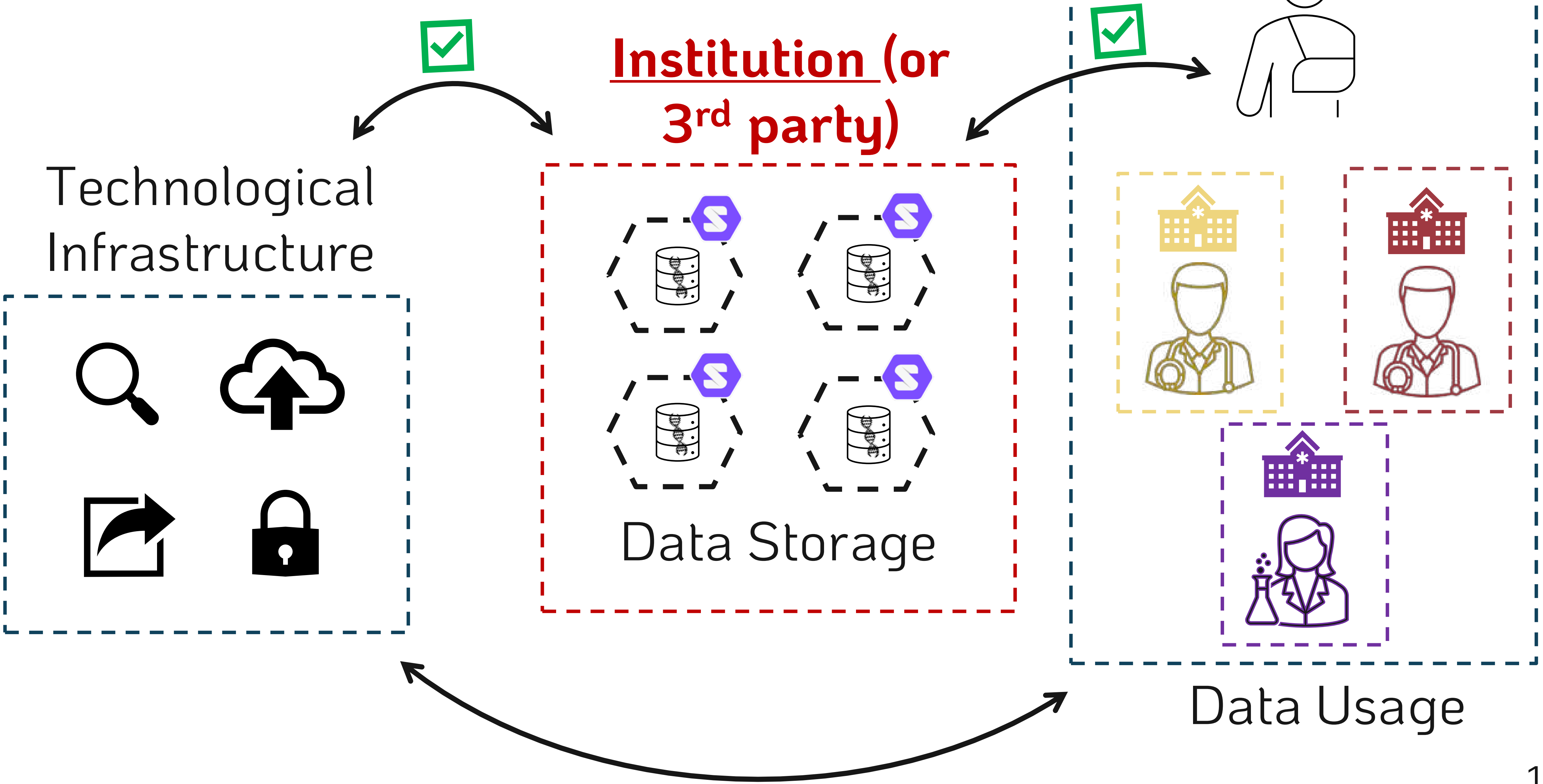
Not much...

Semantic Genome Data?

Institution-centric



A [Hybrid] Solid Solution



Solid Technology

- Does not solve ALL problems, but offers an environment where improvements are possible

  - ✓ Easier sharing (web-facilitated)
  - ✓ Granular privacy controls
  - ✓ Direct patient data access
  - ✓ Data representation flexibility (RDF)
  - ✓ Possibility of data-linking and querying

https://solidproject.org/TR/protocol

Project Goals

Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

While **exploring representation + querying limits**

PhD Goals

✓ Finished
⌛ Soon
Future

Basic Functionalities:

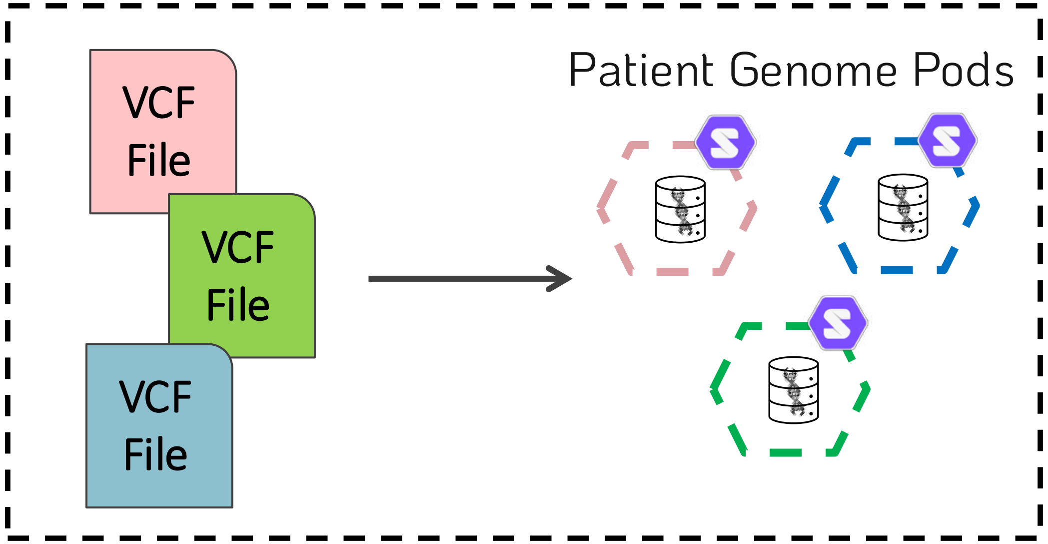
- Genomic data storage in Solid pods ✓
- Data privacy management ⌛
- Data Sharing via the web ✓

Experimental research:

- Represent genomic data as RDF? ⌛
- Linking genomic and health data?
- Single and Many pod Querying?

Workflow Overview

Data Storage



Data Storage

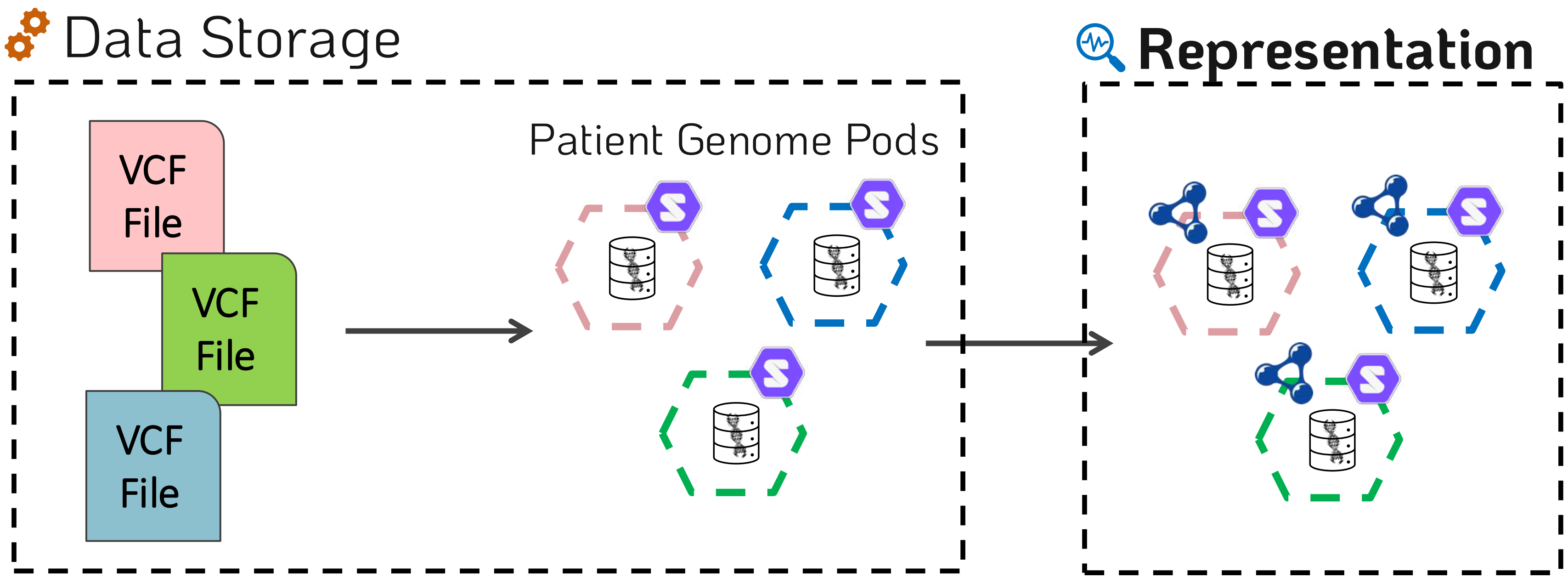
Solid can be implemented for genomic data storage.

Solid pod instances → Community Solid Server[1]

VCF Genomic Data → Illumina Platinum Genomes[2]

[1]https://github.com/CommunitySolidServer/CommunitySolidServer
[2]https://emea.illumina.com/platinumgenomes.html

Workflow Overview

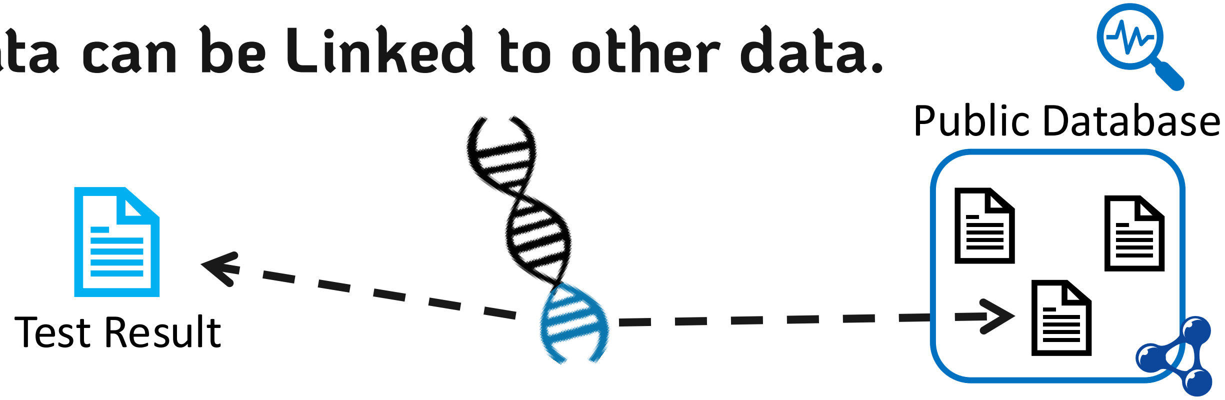


Genome Data in RDF

VCF genomic data can be represented as RDF.

VCF to RDF Vocabularies exist ⟶ SPHN Semantic Interoperability Framework[4] + FHIR HL7 Format[5]

VCF genomic data can be Linked to other data.

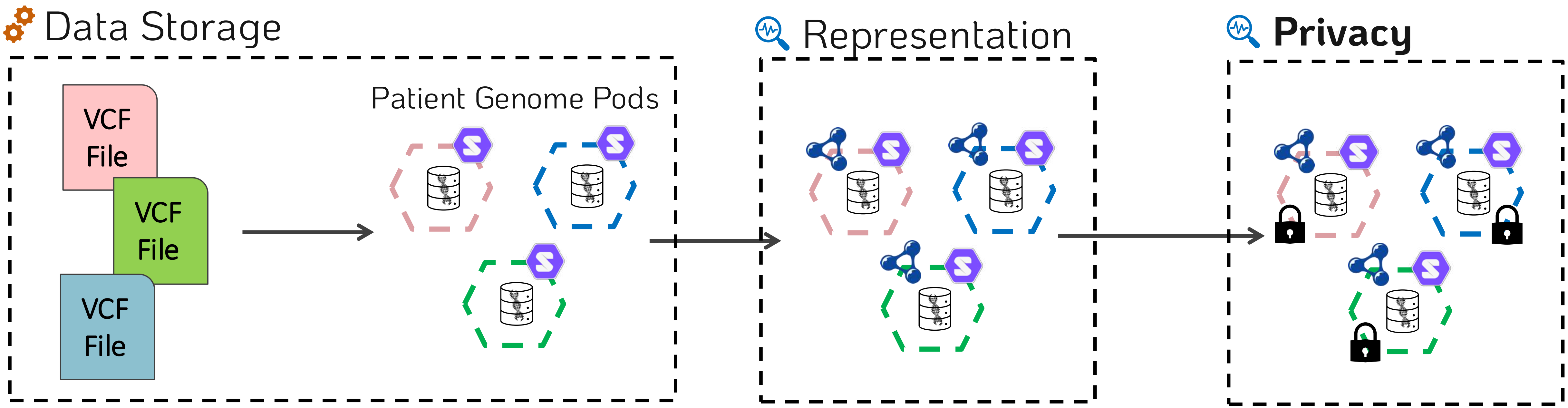


How to automate this?
What vocabularies could be used for the linkages?

[3] van der Horst E, et al. Bridging Clinical and Genomic Knowledge: An Extension of the SPHN RDF Schema for Seamless Integration and FAIRification of Omics Data. 2023.

[4] Dolin RH, et al. vcf2fhir: a utility to convert VCF files into HL7 FHIR format for genomics-EHR integration. 2021.

Workflow Overview



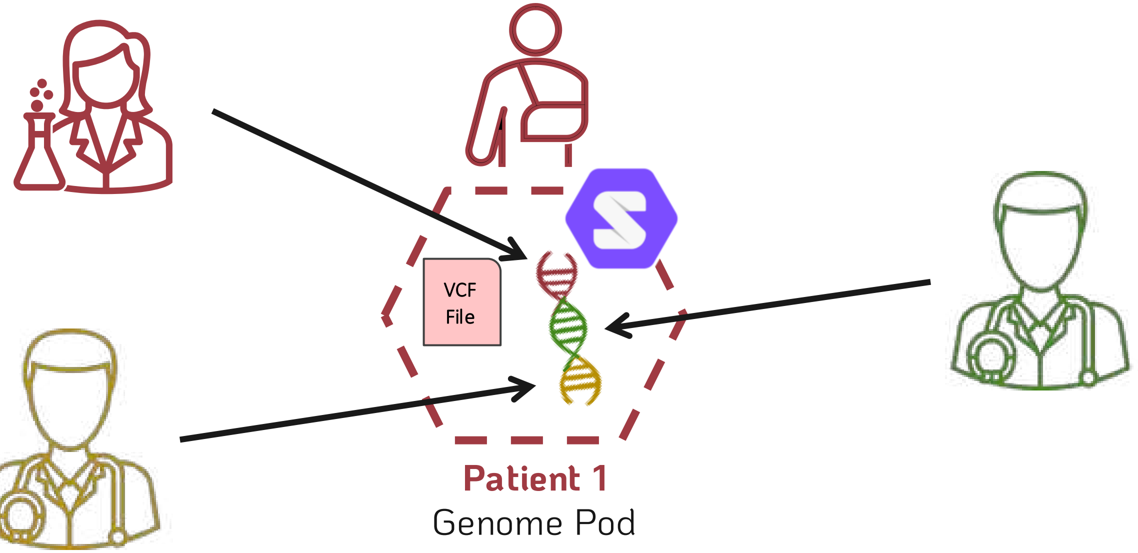
Data Privacy Controls

Granular privacy controls can be implemented in Solid for genomic data.

Web Access Contol → Solid Protocol ACP[3]

How granular can we make the privacy controls?

Privacy controls at chromosome level, or even mutation level.



Chromosome is a semantic Class

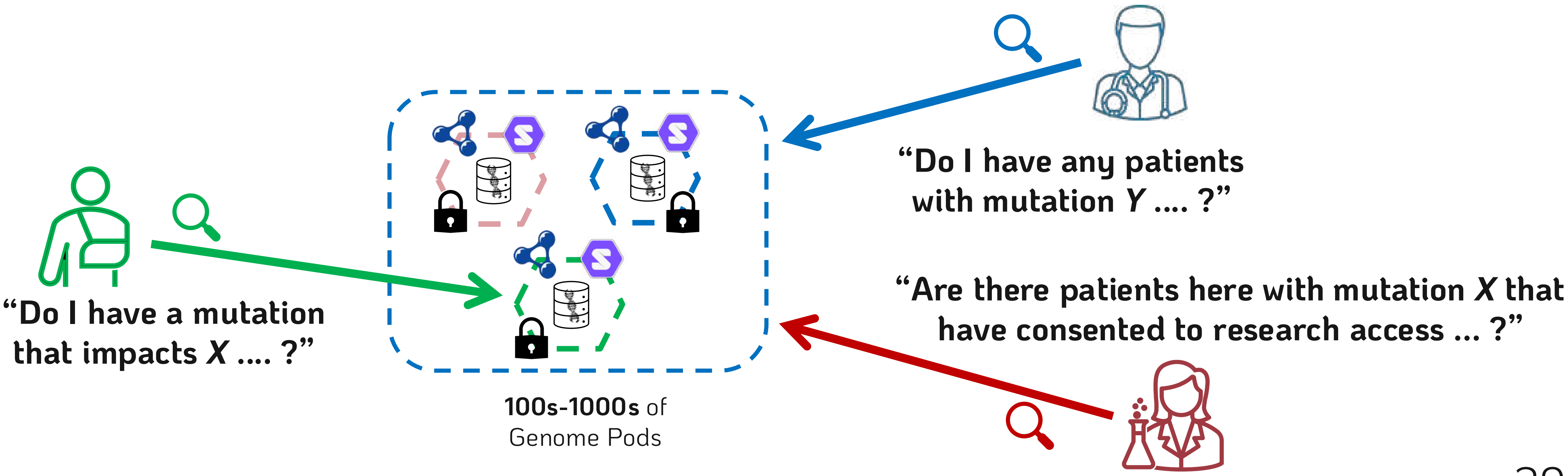
Mutation is also a semantic Class, above a triple but below a Chromosome

[5] https://solidproject.org/TR/protocol

Querying Genome Pods

RDF genomic data pods can be queried.

Query Engine → Custom Comunica implementation/engine[6]



[6] https://comunica.dev/

Querying Genome Pods (possibilities)

Single Pod Query

Utilizes an intra-pod index

Many Pod Query

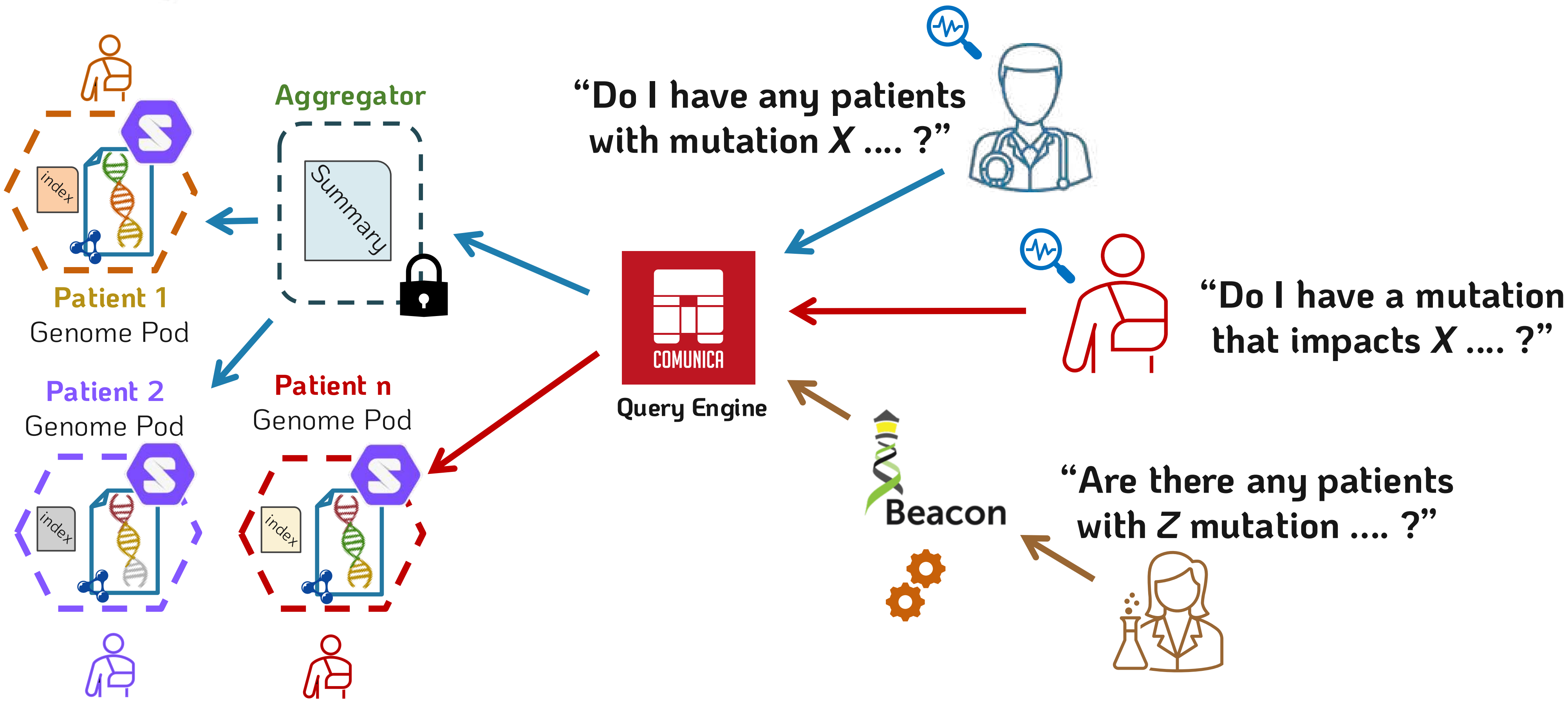
First, utilizes an extra-pod aggregator

Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica

Then, the **Many Pod** query approach is followed



Implementation Specifics

Solid pod instances → Community Solid Server[1]

Genomic Data → Publicly available *VCF* data[2]

RDF Conversion → Aim to use SPHN RDF vocabulary[4]

Data-linking → Actively exploring ontologies / methodologies

Data Querying → Custom instance of Comunica[6]